Thermal management in electric vehicle battery systems

"what are the existing thermal management strategies in conventional battery management systems used in EVs?" *Generated by LeapSpace*

Conventional EV battery thermal management systems (BTMS) mainly use **air cooling, liquid cooling, passive phase-change/heat-storage methods, heat pipes, thermoelectric cooling, and hybrid combinations of these**; among these, **air cooling and liquid cooling are the most established "conventional" approaches**, while PCM and heat pipes are commonly used to improve temperature uniformity or supplement active cooling [1] [2] [3] [4].

Main strategies used in conventional EV BTMS

Strategy	How it works	Typical strengths and limits
Air cooling	Heat is removed by forced or natural convection using air flow across cells or modules [5] [6]	Simple, lightweight, low cost, and widely used, but weak heat-transfer capacity and poorer temperature uniformity under high loads [5] [7] [8]
Liquid cooling	A coolant such as water/glycol or another liquid carries heat away through jackets, cold plates, or direct/indirect channels [9] [10] [11]	Better heat removal, more uniform temperatures, and preferred for high C-rate or high-power packs, but more complex, heavier, costlier, and can raise sealing/leakage concerns [4] [8] [9]
Passive PCM cooling	Phase change materials absorb heat during melting and release it later, buffering temperature spikes [1] [12] [13]	Low/no parasitic power and good temperature buffering, but limited by low thermal conductivity and reduced effectiveness once the PCM approaches full melting [14] [15] [16]
Heat-pipe cooling	Heat pipes move heat efficiently from cells to a remote sink or cold plate [11] [17]	High thermal conductivity and improved uniformity, but often needs support from another cooling stage because the condenser side still needs heat rejection [11] [16]
Thermoelectric cooling	TEC/Peltier modules actively pump heat away from the battery [15] [18] [19]	Precise control and compact form, but relatively high power demand and lower efficiency than fluid-based systems [15] [20]

Strategy	How it works	Typical strengths and limits
Refrigerant/vapor-compression cooling	Battery cooling is integrated with refrigerant loops or chiller-based systems [2] [21] [22]	Strong cooling capability and suitable for integrated vehicle thermal systems, but adds system complexity and energy cost [21] [23]
Hybrid systems	Combine two or more methods, such as PCM + liquid cooling, air + liquid cooling, or heat pipe + PCM [24] [25] [26] [27]	Aim to balance cooling performance, energy use, cost, and temperature uniformity; increasingly favored where single-mode systems are insufficient [24] [25] [28]

What counts as “conventional” in practice

- The literature consistently treats **air cooling and liquid cooling** as the core conventional BTMS families, with **PCM, heat pipes, and TECs** as established alternatives or add-ons [1] [2] [3] [4].
- Reviews also note that **hybrid BTMS** are the main evolution path from conventional single-mode systems because they combine the strengths of active and passive methods [24] [25] [28] [29].
- For high-power EV packs, **liquid cooling is usually the most capable conventional option**, while air cooling remains attractive where simplicity, cost, and low mass matter most [4] [8] [16].

Bottom line

The conventional EV BTMS toolbox is basically a trade-off set: **air cooling for simplicity**, **liquid cooling for performance**, **PCM and heat pipes for passive buffering and uniformity**, and **TECs/refrigerant loops or hybrids when tighter control is needed** [2] [4] [24] [27].

Confidence level: **High** — multiple foundational reviews and more recent reviews consistently describe the same small set of conventional EV battery thermal management strategies and their main trade-offs [1] [2] [3] [4].

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EV battery thermal management strategies

"what about managing temperature in harsher weather conditions like below 15 degree celsius and above 40 degree celsius?" *Generated by LeapSpace*

For **below 15 °C**, the main need is **battery heating/preheating plus insulation and temperature retention**; for **above 40 °C**, the main need is **stronger cooling, often liquid- or hybrid-based**, because battery performance drops in cold and degradation/safety risks rise in heat [1] [30] [31].

What to do below 15 °C vs above 40 °C

Condition	Best-supported thermal response	What the evidence says
Below 15 °C	**Preheating + insulation + heat retention**	Cold temperatures slow electrochemical reaction rates, raise internal resistance, and reduce usable capacity, so BTMSs need heating rather than just cooling [30] [31] [32]. Studies report that **PCM-based systems**, **PCM + heat pipe**, **refrigerant-based heating**, **liquid immersion heating**, and **thermoelectric systems** can warm packs into the operating window [33] [34] [35] [36] [37] [38]. One all-climate system kept batteries **above 20 °C** for hours at subzero ambient temperatures using **PCM + heat pipes + insulation** [34].
Above 40 °C	**High-capacity active cooling, often liquid or hybrid cooling**	High temperatures accelerate aging, increase internal resistance growth, and raise thermal runaway risk, so the priority is to keep the pack below the upper limit and maintain uniformity [31] [39] [40] [41]. Evidence favors **liquid cooling**, **liquid-to-vapor systems**, and **hybrid liquid + PCM or TEC systems** when ambient temperature is high or load is severe [42] [43] [44] [45] [46] [47]. In one hybrid design, active cooling was activated when ambient temperature exceeded **303.15 K** and discharge rate exceeded **5C** [44].

Practical takeaway

- **If the ambient is below 15 °C**, the BTMS should shift toward **heating mode**: preheat to the normal operating band, reduce heat loss with insulation, and use PCM/heat-pipe or TEC-assisted systems when faster warm-up is needed [32] [35] [46] [48].
- **If the ambient is above 40 °C**, the BTMS should shift toward **aggressive cooling mode**: liquid cooling is usually the baseline, but **hybrid systems** are better when the pack sees high discharge rates or sustained heat [42] [43] [44] [47].
- The broader literature still treats the battery's **ideal operating window as roughly 15–40 °C**, so management below and above that range is mainly about getting back into that band and holding temperature uniformity [1] [39] [49].

Bottom line: below 15 °C, you generally need **preheating and heat retention**; above 40 °C, you need **strong active cooling**, with **hybrid BTMSs** offering the most flexible response when weather and load are both harsh [1] [34] [35] [47].

Confidence level: **High —** multiple independent sources consistently support the same cold-weather heating and hot-weather cooling split, with several studies giving direct operating thresholds and examples across both extremes [1] [34] [35] [44] [47].

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Innovation in EV battery thermal management

"so in what direction should an innovator think to take, having these limitations in mind, to implement a solution to manage heat in Li batteries in EVs?" *Generated by LeapSpace*

The strongest direction is to build **hybrid, climate-adaptive, intelligence-assisted BTMS** rather than trying to perfect a single cooling method: combine **liquid cooling for high heat flux**, **PCM/heat pipes for buffering and temperature uniformity**, and **smart sensing + predictive control** so the system can switch behavior across load, ambient, and charging conditions [50] [51] [52] [53] [54] [55].

Where to focus innovation

Direction	Why it is attractive	Main limitation it addresses
Hybrid thermal architecture	Hybrid systems repeatedly show the best balance of peak-temperature control, uniformity, and energy use, especially when they combine passive and active methods [24] [51] [52] [54] [56].	Single-mode air or PCM systems struggle under high-power and extreme-condition use [57] [58] [59].
High-performance liquid cooling	Liquid systems remain the most practical backbone for high-power EV packs and are already the dominant real-world choice in many designs [9] [58] [60].	Needs lower parasitic loss, less mass, and better packaging [60] [61].
PCM + conductivity enhancement	PCMs are strong for passive buffering and temperature smoothing, but need conductivity boosters, encapsulation, and better shape-stabilized designs [59] [62] [63] [64].	Low thermal conductivity and leakage reduce usefulness in fast charge/high load situations [59] [65] [66].
Heat-pipe-assisted designs	Heat pipes improve spreading and hotspot mitigation and work well as a bridge between cells and a remote sink [11] [17] [67].	They still need a condenser-side rejection stage, so they are usually not a standalone answer [11] [67].
AI / model-predictive control	Recent work points to AI-driven control, adaptive MPC, surrogate modeling, and real-time monitoring as key next steps for reducing overshoot and energy use [68] [69] [70] [71] [72].	Static control cannot respond well to rapidly changing driving, charging, and weather conditions [68] [73].

Direction	Why it is attractive	Main limitation it addresses
Extreme-condition design	Innovations should explicitly handle low-temperature heating and high-temperature cooling in the same architecture [1] [53] [55] [73].	Many designs still optimize only one operating band and miss climate robustness [61] [74].

The most promising product direction

An innovator should think in terms of an **integrated thermal platform**, not just a cooler:

1. **Base layer:** efficient **liquid cooling** for routine and high-load heat removal [9] [60] [75].
2. **Buffer layer: PCM or CPCM** to absorb spikes and smooth gradients [59] [62] [64].
3. **Heat-spreading layer: heat pipes or microchannels** to move heat away from hotspots quickly [11] [17] [76].
4. **Brain layer: sensors + AI / MPC / digital-twin style control** to forecast thermal load and adapt flow, fan, or coolant strategy in real time [50] [68] [69] [77] [78].
5. **System-level layer:** co-design with **pack geometry, tab cooling, module layout, and vehicle HVAC integration** to reduce the thermal burden before active cooling has to work [60] [61] [79] [80].

Bottom line

The best innovation path is **not** "one better cooler," but a **modular, hybrid, adaptive BTMS** that is **liquid-cooling-led**, **PCM/heat-pipe-supported**, and **AI-controlled** so it can handle both **high-power fast charging** and **extreme weather** with lower energy penalty, better uniformity, and less thermal runaway risk [52] [54] [55] [64] [81].

Confidence level: **High** — the retrieved reviews and case studies consistently converge on hybrid architectures, improved passive materials, and intelligent control as the main forward direction, while also agreeing on the practical limits of single-mode air, PCM, or passive-only solutions [24] [50] [52] [54] [60].

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Adoption of hybrid thermal management in electric vehicles

"How widely are hybrid thermal management strategies used in EVs currently?" *Generated by LeapSpace*

Hybrid thermal management is **widely researched and increasingly adopted in prototypes and integrated EV thermal systems, but it is not yet the dominant mainstream commercial solution**; the evidence points to **growing use as a high-performance design direction**, while **commercial EVs still rely mostly on liquid cooling, with air cooling in simpler cases and hybrid systems used where performance demands justify added complexity** [51] [60] [82] [83] [84] [85].

How common hybrid strategies are

Setting	What the evidence shows
Research literature	Hybrid BTMS is a major and expanding topic: reviews describe it as the focus of recent work, especially combinations such as **PCM + liquid cooling**, **air + liquid**, **heat pipe + cold plate**, and **TECs with liquid or air systems** [28] [51] [82] [83] [86] [87].
Prototype / advanced system designs	Hybrid systems are repeatedly reported as effective under **fast charging**, **high C-rate discharge**, and **harsh ambient conditions**, often improving temperature uniformity and lowering peak temperature versus single-mode cooling [84] [88] [89] [90] [91] [92].
Commercial reality	The evidence still frames **liquid cooling as the benchmark and most prevalent method** in EV battery thermal management, while direct/hybrid approaches are described as promising but constrained by complexity, cost, and deployment barriers [60] [84] [85] [93] [94].

Bottom line

Hybrid BTMS is common in the engineering literature and increasingly attractive for next-generation EVs, but it is still best viewed as an emerging-to-maturing solution rather than the universal standard in current production EVs [51] [60] [82] [83] [85]. The practical pattern is: **single-mode liquid cooling remains the default, and hybrid architectures are chosen when the pack must handle higher thermal loads, faster charging, tighter temperature uniformity, or extreme climates** [84] [88] [90] [92] [95].

Confidence level: **High** — multiple reviews and application studies consistently show strong momentum for hybrid BTMS, while the same evidence also indicates that commercial deployment still favors simpler liquid-based systems [51] [60] [82] [84] [85].

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Integration of active cooling in hybrid battery thermal management

"How well can active cooling systems in the hybrid thermal management strategies be integrated into the designs of the EVs?" *Generated by LeapSpace*

Active cooling integrates well into EV designs when it is treated as a vehicle-level subsystem, not just a battery add-on. The strongest pattern in the evidence is that **liquid- or refrigerant-based active cooling can be packaged into existing EV thermal loops, battery enclosures, and HVAC architectures**, especially when paired with **smart control, modular pack segmentation, and hybrid passive elements** to manage heat, weight, and energy use [60] [96] [97] [98].

How well integration works

Integration path	What the evidence shows	Design implication
Battery-side direct or indirect liquid cooling	Integrated liquid cooling systems can hold battery temperature in safe ranges under high ambient heat, and control performance improves when pump, compressor, and valves are co-designed with the pack and HVAC loop [98] [99] [100].	This is the most practical backbone for mainstream EVs because it fits existing vehicle thermal circuits [60] [100].
Direct refrigerant / vapor-compression cooling	Direct cooling systems can achieve tight temperature control, with one integrated system limiting fluctuations to less than 0.7 °C and another supporting dual-mode cooling/heating with full-pack retrofitting potential into liquid-cooled architectures [96] [101].	Best suited where compactness and high heat rejection matter, but it needs careful control because refrigerant coupling is strong [96] [101].
Hybrid battery + cabin thermal management	Multiple studies show that sharing hardware between cabin and battery can improve space efficiency and energy use, while maintaining battery temperatures in safe bands even at high ambient temperatures [97] [98] [102] [103].	Integration works best when the battery is prioritized thermally but the cabin loop is still coordinated through control logic [97] [104] [105].
Heat-pipe and PCM-supported active cooling	Heat pipes, PCM, and delayed liquid cooling reduce peak temperature and temperature spread while lowering reliance on active cooling during moderate loads [89] [106] [107] [108].	A hybrid stack can shrink the active cooling burden, which helps packaging and parasitic-power constraints [107] [108].

What makes integration succeed

- **System-level co-design** is the key enabler: the strongest studies embed the battery cooler, HVAC, compressor, pumps, valves, and control logic in one unified thermal architecture rather than optimizing each part separately [98] [102] [109].
- **Control quality matters as much as hardware**: decoupling control, gain-scheduling, fuzzy logic, MPC, and flow-distribution strategies are repeatedly used to keep battery and cabin temperatures on target while limiting energy penalty [96] [98] [104] [110] [111].
- **Packaging trade-offs remain real**: advanced indirect systems can improve performance and safety, but they often add weight, complexity, and integration burden unless the design is compact and modular [54] [98] [112].
- **Retrofit potential is strong for liquid-cooled platforms**: direct refrigerant cooling studies explicitly note that existing liquid-cooled pack architectures can be adapted with limited battery-side modification [101].

Bottom line

Active cooling integrates well into EVs when it is built into the vehicle's shared thermal architecture and controlled adaptively. In practice, **liquid cooling is the easiest integration baseline**, **direct refrigerant cooling is highly capable but more control-sensitive**, and **hybrid designs with PCM or heat pipes are the best route when the goal is to reduce energy use, improve uniformity, and keep the package compact** [54] [60] [96] [98] [107].

Confidence level: **High** — multiple independent studies and reviews consistently show that active cooling is feasible and effective in EVs, especially when integrated with HVAC and vehicle-level control rather than implemented as a standalone battery cooler [60] [96] [98] [102].

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